

Perforce P4 Runbook - Alert Handling Procedures

Perforce Professional Services

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Preface

These are detailed "runbook" procedures for P4 tasks involved with handling alerts.

The intention of this doc is that anchors are provided for each alert type, so that the alert message can link directly to the relevant section of this document. This is particularly useful for Slack notifications etc.

This version is intended to be generic and for organisations to use as a template for their own alert handling procedures. It is not intended to be a definitive guide for all alert handling, as each organisation will have their own processes and procedures.

Examples alerts are in [Example perforce alert rules](#).

If an alert has the attribute `runbook_id` then this is the anchor to link to in this document for the alert handling procedures.

```
- alert: Replication Errors
  # This means we have a p4 pull value of -1 or similar
  expr: p4_pull_replication_error > 0
  for: 30m
  labels:
    severity: "high"
  annotations:
    summary: 'Endpoint {{ $labels.instance }} replication error occurring'
    description: "{{ $labels.instance }} of job {{ $labels.job }} has been true for 30m."
  runbook_id: _replication_error
```

Chapter 1. Basic workflow for alert handling

1. Monitor your Slack channel for new alerts:

There are links on the bottom row to this document, and the node exporter and p4prometheus (if relevant) dashboards in Grafana for the alerting instance.

Examples of Slack alert templates in [Example Slack alert templates](#)

2. Review severity and respond with appropriate urgency (both investigating and responding to users as appropriate)
3. Investigate alert and perform resolution steps
4. Escalate to other teams if necessary (e.g. storage, networking, etc)
5. Improve Alert Handling Procedure
 - Evaluate incident
 - Did you find a bug?
 - Did you learn something new worth sharing?
 - Update documents
 - Share learnings, and suggest opportunities for improvement

Chapter 2. Severity 1 Alerts

2.1. P4D service not running

Verify on server and try to identify cause of failure. In example commands below, the assumption is instance **1** is used - adjust if necessary.

- Is service running?

```
ps aux | grep p4d
p4 info
sudo systemctl status p4d_1
```

- User stopped service? E.g. **p4 admin stop**

```
tail -1000 /p4/1/logs/log | less
```

- Errors in p4d log?

```
grep -A5 error: /p4/1/logs/log | less
```

- Service died?

```
sudo journalctl -u p4d_1
sudo journalctl -u p4d_1 --no-pager | less
```

- OOM or other **syslog** errors?

```
sudo grep oom /var/log/messages
sudo grep p4d /var/log/messages
```

Resolution

- Fix problem identified above and restart P4D!

```
sudo systemctl restart p4d_1
ps aux | grep p4d
p4 info
```

2.2. P4D SSL certificate expiry

This alert indicates the SSL certificate has expired

Resolution

- Re-generate p4d certificate - **IMPORTANT** all users will have to run `p4 trust -yf` or equivalent again, including bots, replicas etc. So this will require communication with users.

```
p4d -Gf
# Examine the output of the above
p4d -Gc
# Optional - remove existing certs
cd /p4/ssl
mkdir save
mv * save/
p4d -Gc          # Generate new certificate
p4d -Gf          # Validate new certificate
p4 trust -yf     # Update trust on server
```

Optional: use organization provided certificate instead of self-generated one.

2.3. Swarm unhealthy

This alert indicates that Swarm is not responding to requests from commit server.

Recent occurrences have been due to memory issues with `php-fpm` service.

Resolution

- Look at node_exporter dashboard to check for CPU/memory
- Login to Swarm box and check status of `httpd` (Rocky) or `apache2` (Ubuntu) service
- Review `atop` and historical data (`sudo atop -r /var/log/atop/atop_20250125` or similar)
- Look for php-fpm errors (used up all memory), and restart the service: `systemctl restart php-fpm`
- Look for errors in `/var/log/messages`
- Look for errors in `/opt/perforce/swarm/data/log`
- Try running `curl` to Swarm URL (from `p4 property -l` on commit server)
- Make sure DNS is correctly setup

2.4. Diskspace Below Filesys Config - P4D STOPPED!

This is a generic error for all filesystems - specifics are noted below for `/hxlogs`, `/hxmetadata` and `/hxdepots`.

Note that as far as users are concerned `p4d` is not working when this error occurs, as the user will get an error message whatever command they run (except for some reporting commands such as `p4 info`)

Basic actions are:

1. Check values of volumes and relevant configurables:

```
df -h
df -h | grep /hx
p4 configure show | grep filesys
p4d -cshow | grep filesys      # Assumes p4_vars has been sourced
```

Resolution

- Review dispace trends in Grafana dashboard for node exporter (follow link in slack alert message)
- Check for any large/unexpected files, eg. in `ls -lhrS /p4/1/logs/ | tail` and consider deleting them (**but not log or journal!!!!**)
- Temporarily decrease relevant configurable (see next sections)
- Follow local processes to increase volume size and grow filesystem if appropriate

2.5. Diskspace Below Filesys Config /hxlogs - P4D STOPPED!

See common issues [Section 2.4, “Diskspace Below Filesys Config - P4D STOPPED!”](#). For this volume there may be:

- Unexpectedly large `journal` or `log` in `/p4/1/logs/`

Resolution

- Consider general resolution steps as above [Section 2.4, “Diskspace Below Filesys Config - P4D STOPPED!”](#)
- Configure the values in `p4metrics.yaml` to auto-rotate `journal` and `log` if they are too large (see [example p4metrics.yaml](#) for details)
- Temporarily reduce configurable - this can't be done using front door `p4 configure set` due to the error:

```
p4d -cshow | grep filesys
# Set appropriate ones, e.g. reduce from 5G to 3G
p4d "-cset filesys.P4JOURNAL.min=3G"
p4d "-cset filesys.P4LOG.min=3G"
# Cross check again
p4d -cshow | grep filesys
# Make sure server commands are going through again
p4 depots
```

- Immediately rotate journal if it is very large (will move it to `/hxdepots`) - **NOTE this will block server while being moved and it can take a few minutes for large journal as this involves a copy between filesystems:**

```
p4 admin journal
```

- If `/p4/1/logs/journal` is regularly growing very fast then configure `p4metrics.yaml` as above to rotate it more regularly or at specific thresholds
 1. P4metrics will collect journals metrics in which tables are being used `p4_journal_records_count` - look for particularly large values for specific tables
 2. Also possible to analyse a journal to see which tables are being used:

```
cd /p4/1/checkpoints
# Pick appropriate rotated journal number, e.g. 793
nohup cat p4_1.jnl.793 | grep --color=auto -E "^@(dv|rv|pv)" | cut -d' ' -f 1,3 |
sort | uniq -c > tables.txt &
```

2.6. Diskspace Below Filesys Config /hxmetadata - P4D STOPPED!

1. Check sizes of root/offline_db

```
du -sh /p4/1/root/
du -sh /p4/1/offline_db/
```

Resolution

- Most likely you will need to increase volume size as above [Section 2.4, “Diskspace Below Filesys Config - P4D STOPPED!”](#)
- Temporarily reduce configurable - this can't be done using front door `p4 configure set` due to the error:

```
p4d -cshow | grep filesystem
# Set appropriate ones, e.g. reduce from 5G to 3G
p4d "-cset filesystem.P4ROOT.min=3G"
# Cross check again
p4d -cshow | grep filesystem
# Make sure server commands are going through again
p4 depots
```

- If `offline_db` is much smaller than `root`, then consider swapping (consult first - there is an SDP script)

2.7. Diskspace Below Filesys Config /hxdepots - P4D STOPPED!

1. Check values as above [Section 2.4, “Diskspace Below Filesys Config - P4D STOPPED!”](#)

Resolution

- Check Grafana for recent disk usage trends
- Most likely this will require `/hxdepots` volume to be expanded - see above [Section 2.4, “Diskspace Below Filesys Config - P4D STOPPED!”](#)
- Review the size of `/p4/1/checkpoints/` and consider either deleting older journals/checkpoints if they are large or if there are lots of them:

```
du -sh /p4/1/checkpoints/
ls -ltrh /p4/1/checkpoints/
```

- It **might** be an idea to reduce the value of the `KEEP` variables (so that not as many are retained as currently if taking up too much space):

```
grep KEEP /p4/common/config/p4_1.vars
```

- Temporarily reduce configurable - this can't be done using front door `p4 configure set` due to the error:

```
p4d -cshow | grep filesystems
# Set appropriate ones, e.g. reduce from 5G to 3G
p4d "-cset filesystems.depot.min=3G"
# Cross check again
p4d -cshow | grep filesystems
# Make sure server commands are going through again
p4 depots
```

2.8. Diskspace Low - Proxy STOPPED!

1. Check values as above [Section 2.4, “Diskspace Below Filesys Config - P4D STOPPED!”](#)

Resolution

- Check Grafana for recent disk usage trends
- Proxy usage when there is a new release can quite quickly overload the cache filesystem
- This most likely is an instance where there is proxy cache cleaner script running regularly
- Run the script to reduce load immediately
- Consider running it more frequently (in the crontab)
- Quite likely this will require `/hxdepots` volume to be expanded - see above [Section 2.4,](#)

“Diskspace Below Filesys Config - P4D STOPPED!”

```
du -sh /hxdepots/
```

2.9. Replication Errors

This indicates an error from `p4 pull -ljb` (don't use `-v` on p4d servers < 2021.1) on the replica. This is important to investigate if it is an HA replica.

Resolution

- Login to instance
- Restart p4d if required
- Identify any replication errors and if necessary fix/restart
- Temporarily reduce configurable - this can't be done using front door `p4 configure set` due to the error:

```
p4 pull -ljb
# The -v option supported p4d 2021.1+
```

- Check the SDP resolution see [Replication appears stalled](#)

In case of stuck replication:

```
$ p4 pull -ljb
Current replica journal state is:      Journal 3443,   Sequence 344962308.
Current master journal state is:      Journal 3443,   Sequence 841624005.
Current master journal state is:      Journal 3444,   Sequence 137848798.
Replica is 634510152 bytes of journal behind the upstream server (1 journal rotations)
The statefile was last modified at:   2025/12/12 15:28:46.
The replica server time is currently:  2025/12/13 23:39:36 +0000 UTC

perforce@p4-edge:/home/perforce p4 monitor show -al
1014510 B svc_p4d_edge_fra 23:28:04 pull -i 1 Journal file '-' replay failed at line
1!
Bad opcode '1765553439' journal record!
```

This indicates a corrupt journal record (can happen when upstream server is forcibly shut down). To resolve:

- Stop the replica p4d service
- Move the current pull state file to a backup location
- Restart the replica p4d service
- Check status of replication again

```
sudo systemctl stop p4d_1
cd $P4ROOT
mv pull_state save/
sudo systemctl start p4d_1
p4 pull -lrv
p4 monitor show -al
```

2.10. Pull Queue Stuck

This alert fires when the replica pull queue has exceeded 10,000 files for over 10 hours.

This can happen for a variety of reasons, most commonly:

- Non-existent (purged) files (where filetype includes +Sn - where n is number of revisions to keep contents for)
- Non-existent (shelved) files
- Non-existent files with verify problem on master server
- Temporary file transfer problems which exceeded thresholds for auto-retry

Resolution

- Check pull queue contents:

```
p4 -e pull -l | head -20
```

- Retry stuck files

```
p4 pull -R
```

- Verify that the files transferred:

```
p4 pull -ls
```

2.11. Pint problem

This is a problem with the Pint service, which is used to monitor the health of the prometheus service and our Alertmanager rules.

Pint is the Prometheus linter, which checks the Prometheus rules and alerts for errors - see <https://github.com/cloudflare/pint> for details.

There are 2 main types of problem when this alert happens:

1. There is a typo in the Prometheus rules file, which causes Pint to fail

2. There is a problem with the prometheus data which the rules are querying.

Resolution

The first is the most common, and can be fixed by correcting the rules file and running `make` to validate the rules and redeploy them.

```
# Login the HRA Monitoring server
sudo su - prometheus
cd /etc/prometheus
# Validate the rules
make
# Fix any errors reported by Pint
# If no errors, then deploy the rules
make deploy
```

The second is more complex, and may require investigation of the Prometheus data to see if it is valid.

- Use Grafana Explore option, to query `pint_problem` metric - this will indicate which rule is causing the problem.
- Check the Prometheus rules file for the rule that is causing the problem and copy the query to the Grafana Explore to find out what is happening.
 - It is usually a problem with data being somehow duplicated or 2 time series overlapping
 - Examples include:
 - the mountpoint for `/hxdepots` has changed - so there are 2 label series for the same mountpoint but with different values for `device` label
- Resolution is typically to try deleting the group which is causing the problem.
- Check the query in Grafana Explore again to ensure it is now working correctly (adjust time interval, e.g. to 5m or less to exclude older data)

Chapter 3. Severity 2 Alerts

3.1. P4D urgent license expiry

Validate the alert with Grafana and the server.

Resolution

Coordinate with licensing team to get new license.

3.2. P4D license data missing

Validate the alert with Grafana and the server.

Resolution

Coordinate with licensing team to get new license.

3.3. P4prometheus service not running

Verify on server. Note reason for failure and restart.

Resolution

- Ensure `server` ≥ 3 and `track` = 1:

```
p4 configure show track
p4 configure show server
p4 configure set track=1
p4 configure set server=3
```

- Restart p4prometheus:

```
sudo systemctl restart p4prometheus
```

3.4. P4metrics service not running

Verify on server. Note reason for failure and restart.

Resolution

- Review logs:

```
sudo journalctl -u p4metrics
```

- Ensure `p4metrics` version is recent enough: `v0.10.0` or later

```
p4metrics --version
p4metrics, version v0.10.0 (branch: master, revision: 1daae6f)
  build user:      rcowham@perforce.com
  build date:      2025-04-16T13:56:17+0100
  go version:      go1.24.2
```

- Restart p4metrics:

```
sudo systemctl restart p4metrics
```

3.5. Missing Checkpoint Metrics

This alert fires when a Perforce server (**commit-server**, **standard**, or **edge**) does not expose any checkpoint metrics for at least 5 minutes.

It is intended to detect issues where:

- No checkpoints are running
- Monitoring is misconfigured
- The server is not emitting the **p4_sdp_checkpoint_duration** or **p4_sdp_checkpoint_log_time** metrics

Resolution

- Verify **node_exporter** is running on the P4D server:

```
systemctl status node_exporter
```

- Check checkpoint metrics

```
curl -s localhost:9100/metrics | grep checkpoint
```

You should see these two metrics

```
p4_sdp_checkpoint_duration
p4_sdp_checkpoint_log_time
```

- Verify the **.prom** files exist on the server:

```
ls -l /hxlogs/metrics/p4_checkpoint-*.prom
```

The timestamp should be recent

- Review installation of monitoring

3.6. Replication slow

This alert indicates that replication pull queue (metadata) is slow.

Review past history

Resolution

1. If the pull queue is advancing, even if slow then check networking or other issues
2. See [Section 2.9, “Replication Errors”](#) for detailed steps, especially if it appears to be stalled.

3.7. Network access problems

Initial basic questions to ask:

1. When did this start happening?
2. Has anything changed?
3. Is it affecting all users or just some users?

Resolution

1. Confirm p4d service(s) running on the VM
2. Review server log to see if any ongoing or recent access:

```
tail -f /p4/1/logs/log
p4 monitor show -al
```

3. Confirm local firewall rules are not blocking p4d port
4. If you have access to AWS security group or Azure networking console, then confirm access rules look OK (check to see if any recent changes)
5. Test access from another client machine if possible

3.8. Network performance problems (slow syncs or submits)

Initial basic questions to ask:

1. When did this start happening?
2. Has anything changed?
3. Is it affecting all users or just some users?

Resolution

1. Review p4d logs for errors:

```
less /p4/1/logs/errors.csv
```

```
grep -A4 error: /p4/1/logs/log | less
```

2. Check if the following configurables are set (these relate to large syncs or submits and users reporting disconnects):

```
net.keepalive.idle: 180
net.keepalive.interval: 20
net.keepalive.count: 6
net.autotune=0 <== P4D 2022.2 and earlier versions only
net.parallel.threads=4
net.parallel.sub.threads=4
net.parallel.submit.batch=8
net.parallel.submit.min=4
```

For a user with a slow Internet connection, set `net.maxwait=60'` either for the server (via p4 configure) or in the workspace's `.p4config` file. Setting it for the server requires a stop and start (NOT restart) of the p4d service.

3.9. Node exporter parse error detected

This alert means that `node_exporter` is not able to parse the metrics from the server - the files in `/p4/metrics/*.prom`.

Most likely this is due to a label formatting problem.

Resolution

Upgrade to the latest version of `p4metrics` (v0.11.0 or later) as it sanitizes labels better.

Login to the server and review issues with the `node_exporter` service:

```
sudo systemctl status node_exporter
sudo journalctl -u node_exporter --no-pager | less
```

You may need to restart the `p4prometheus` service:

```
sudo systemctl restart p4prometheus
sudo systemctl status p4prometheus
```

3.10. OOM Kill Detected

OOM (Out of Memory) kill detected means that the operating system has killed a process (in this case likely `p4d`) because it was using too much memory. It has also been observed many times on a Swarm/Code Review server, where the `php-fpm` service is using up all the memory and causing the OOM killer to kill processes.

- Check Grafana and investigate on server using atop and logs to try to identify cause.
- Check instance type and available memory.

Resolution

- See [Section 2.3, “Swarm unhealthy”](#) for resolution steps if this is on a Swarm/Code Review server.
- User education is important to avoid this happening again - consider reaching out to users to explain what happened and how to avoid it in the future
- Might consider upgrade of instance type for more memory if this is a common occurrence, but this should be a last resort after user education and other mitigations such as setting up swap space.

3.11. OOM P4Metrics Issue Detected

This is very similar to the above ([Section 3.10, “OOM Kill Detected”](#)), but it is the `p4metrics` service which is killing the process (or terminating it) due to OOM thresholds exceeded.

Examples of this behaviour are often inexperienced users who attempt:

- a sync of a workspace which includes tens of TB/tens of millions of files - because they are trying to sync a depot with many streams/branches rather than a single one
- reporting commands trying to run on a very large number of files, e.g. `p4 files //depot/...` or `p4 filelog //depot/...`
- a "p4 print" on a large number of files
- a user ending up with many "active" processes, often due to tool usage (accidentally keeping connections open), which even if in Idle state consume lots of RAM (e.g. 20-30 processes each using 3% of RAM)

Resolution

Investigate which command was terminated and why (it will be documented in the journal logs for the `p4metrics` service) and then take appropriate action.

Check the configuration of `p4metrics` to see if the thresholds for OOM are set appropriately for the instance type and workload.

```
/p4/common/config/p4metrics.yaml
sudo journalctl -u p4metrics --no-pager | grep Memlimit | less
```

Chapter 4. Severity 3 Alerts

4.1. Checkpoint Not Taken

This alert indicates a checkpoint may not have been taken.

Examine latest checkpoint log for errors:

```
less /p4/1/logs/checkpoint.log
```

Validate the dates and times of the metrics files (ensure they are all recently updated!):

```
cd /p4/metrics/  
ls -ltr
```

Make sure the **p4metrics** service is running correctly:

```
sudo systemctl status p4metrics
```

Resolution

- Potentially restart **p4metrics** service (if stopped):

```
sudo systemctl restart p4metrics  
sudo systemctl status p4metrics
```

- Resolve any issues with **verify_sdp.sh** for example (will be documented in the log)

4.2. P4D license expiry

Validate the alert with Grafana and the server.

Resolution

Coordinate with licensing team to get new license.

4.3. CPU Usage High

- Check Grafana and investigate on server using **atop** (view historical snapshots every 10 mins) and logs to try to identify cause.
- Check instance type and available memory.
- Consider log analysis to identify culprit(s)
- Consult with users as to any jobs they are running, or known load.

Resolution

Resolutions include setting up swap space or upgrading instance type.

- Ensure THP not enabled - see [SDP docs](#)

or

- Upgrade instance type to one with more CPU cores

4.4. App Memory Usage High

- Check Grafana and investigate on server using atop and logs to try to identify cause.
- Check instance type and available memory.

Resolutions include setting up swap space or upgrading instance type to get more RAM.

Resolution

- Upgrade instance type
- If p4d >= 2023.1 then we should be setting the resource protection options on, and in particular we should make sure that the appropriate capability is enabled.

```
# For case sensitive servers:
getcap $(readlink -f /p4/1/bin/p4d_1)
# For case insensitive servers:
getcap $(readlink -f /p4/common/bin/p4d_1_bin)

# Output should be something like:
/hxdepots/p4/common/bin/p4d_2023.2.1234567 cap_sys_resource=ep

# If required, then set the capability
# For case sensitive servers:
sudo setcap CAP_SYS_RESOURCE=+ep $(readlink -f /p4/1/bin/p4d_1)
# For case insensitive servers:
sudo setcap CAP_SYS_RESOURCE=+ep $(readlink -f /p4/common/bin/p4d_1_bin)

# And re-run the appropriate getcap command to ensure all OK
```

- **Restart p4d_1 service if you had to change the capability**

4.5. HX Diskspace Predicted Low

Verify in Grafana and also on the server. These alerts may be overly cautious, so the algorithm may need to be adjusted to make it less sensitive.

Resolution

- Check if the alert should be tweaked to be less sensitive.

4.6. Root Diskspace Predicted Low

Verify in Grafana and also on the server. These alerts are usually incorrect, so the algorithm may need to be adjusted.

Resolution

- Check if the alert should be tweaked to be less sensitive.

4.7. Diskspace Percentage Used Above Percentage Threshold

This alert indicates disk space usage has crossed a threshold over time.

Resolution

- increase volume size and grow filesystem

Chapter 5. Severity 4 Alerts

5.1. Hansoft Diskspace Low

We shouldn't see this alert, because we have cleanup script running.

- AlertConditions
 - Disk space usage is projected to go below threshold within 48 hours based on current trend
 - Disk space is below 10GB

Resolution

- Check current disk space and compare against alert to confirm validity
- Check status of "clean up script" to ensure its running
- Adjust disk space appropriately

```
df -Th                # check disk usage on all file systems
df -Th | grep /dev/nev # limit usage to just
```